

EXPOSCAN SENIOR FULL STACK ENGINEER

Technical Screening

Created by Sahil Bahety (Founder)

Stack
FastAPI | React 19 | MongoDB | Python

Deadline
5 days from receipt

Submit to
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BEFORE YOU START

This is a short test. Four questions. No boilerplate - we want to understand how you actually think and build. You will be asked to walk through your answers on a call, so write things you can defend.

- All code must be your own. AI-assisted boilerplate you cannot explain will be disqualified.
- For code questions: paste directly or link a GitHub repo with a README.
- Be direct. Longer is not better.

Question 1 - The Build

Tell us about one production system you have personally owned end to end in the last two years.

Cover these four points:

- What was the problem it solved?
- What did you build and what were the key technical decisions?
- Was it multi-tenant? If yes, how did you handle data isolation?
- What broke in production and how did you fix it?

We are looking for: real production experience, architectural ownership, and honest self-assessment. If nothing broke, we do not believe you.

Question 2 - The Architecture

You are migrating ExpoScan - a single-tenant FastAPI + MongoDB PWA - to a multi-tenant SaaS product. Multiple clients will use the same deployment. Their data must be completely isolated.

Answer the following:

- Pick one data isolation approach (shared collection with tenantId, or database-per-tenant) and justify it for this specific context - a trade show lead capture app expecting 50 to 200 tenants at launch.
- Write the FastAPI middleware or dependency that enforces tenant isolation on every request. Show the code.
- What is the single biggest risk in this migration and how do you mitigate it?

We are looking for: real architectural judgment, not a textbook comparison of both approaches.

Question 3 - The Debugging

A FastAPI endpoint that has been running fine for three months suddenly starts returning 500 errors in production. The logs show a MongoDB timeout. No schema changes were made. No code was deployed. Traffic is normal.

Walk through your debugging process:

- What are the first three things you check and in what order?
- What commands or tools do you actually run?
- What is the most likely cause and how do you confirm it?

We are looking for: specificity. 'Check the logs' is not an answer. Which logs, which commands, what patterns.

Question 4 - The Judgment Call

You join as the sole engineer on a live product. In your first week you review the codebase and find three things: (1) no indexes on the leads collection which has 80,000 documents and is growing, (2) the Idempotency-Key check is not atomic - there is a race condition, (3) created_at timestamps are being set by the client, not the server.

Answer these:

- Which do you fix first and why?
- Write the fix for the one you consider most critical.
- How do you communicate these findings to a non-technical founder without causing panic?

We are looking for: prioritisation under real constraints, not a list of everything that needs fixing.

CANDIDATE DETAILS

Candidate Name: _____

GitHub: _____

Years of Experience: _____

Date Submitted: _____